

EECS 16A Foundations of Signals, Systems, & Information Processing

Fall 2025 Homework 9

This homework is due Friday, October 31, 2025, at 23:59.

Submission Format

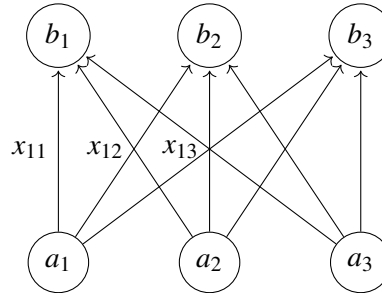
Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned). as well as your IPython notebook saved as a PDF.

On occasion, a problem set contains one or more problems designated as optional. Your solutions to such problems are not to be evaluated, and will not count toward your grade. Nevertheless, you are still responsible for learning the subject matter within their scope. In other words, we may test you on their scope.

1. Linear Neural Networks

Considered the pride and joy of modern-day machine learning, neural networks are actually simply the composition of multiple mathematical functions in order to approximate more complex behavior. In this problem, we will consider a very basic type of neural network, a *fully-connected linear neural network*, and derive closed-form solutions that let us train and perform inference on this machine learning model.

First, we will consider a single-layer neural network, in which a *linear combination* of the input values $a_1, a_2, \dots, a_n$ are used to produce prediction values $b_1, b_2, \dots, b_m$. An example of this architecture for $n = m = 3$ can be seen in the following figure:



In this case, we express mathematically that

$$b_j = \sum_{i=1}^n x_{ij} a_i$$

where each x_{ij} is the *weight* of the connection from input a_i to output b_j (note that only the weights from input a_1 are pictured for visual reasons, but all input-output pairs should have appropriate weights). These model weights are the values that we "learn" from training examples in order to produce a model that can predict new outputs from inputs in the future.

- Using the mathematical definition provided, express b_j as the inner product between two vectors. Be sure to specify the entries of your vectors, and what they represent.

Solution: The summation form $b_j = \sum_{i=1}^n x_{ij}a_i$ can be pattern-matched to the definition of the inner product between vectors the vectors

$$\mathbf{x}_j = \begin{bmatrix} x_{1j} \\ x_{2j} \\ \vdots \\ x_{nj} \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

Therefore, $b_j = \sum_{i=1}^n x_{ij}a_i = \mathbf{x}_j^\top \mathbf{a} = \langle \mathbf{x}_j, \mathbf{a} \rangle$.

Here, $\mathbf{a}$ represents all of our inputs to the neural network, while $\mathbf{x}_j$ represents the weights of each of the inputs to b_j .

- (b) Let us define the vector $\mathbf{b} = [b_1 \ b_2 \ \cdots \ b_m]^\top$ (notice that this is a column vector). Using the inner product you derived above, express $\mathbf{b}$ as a matrix-vector multiplication. Be sure to specify the entries of your matrix and vector respectively, and describe what they represent. Note that this expression actually describes how we perform *inference* (prediction of $\mathbf{b}$ given our inputs) using our neural network.

Solution: We derive in the previous subpart that $b_j = \mathbf{x}_j^\top \mathbf{a}$. For vectors

$$\mathbf{x}_j = \begin{bmatrix} x_{1j} \\ x_{2j} \\ \vdots \\ x_{nj} \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

In order to produce an expression for $\mathbf{b}$, we can "stack" our inner products (observing that $\mathbf{a}$ is the same for each b_j) to get

$$\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} = \begin{bmatrix} x_{11} & x_{21} & \cdots & x_{n1} \\ x_{12} & x_{22} & \cdots & x_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1m} & x_{2m} & \cdots & x_{nm} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

If we define

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1m} \\ x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

then $\mathbf{b} = \mathbf{X}^\top \mathbf{a}$.

As with the previous subpart, $\mathbf{a}$ represents all of our inputs to the neural network. The matrix $\mathbf{X}$ now represents all of our learned model weights x_{ij} .

If we actually want to use our neural network architecture, we first need to train it! In order to do this, we will estimate the weights x_{ij} in our network that best approximate the desired output $\mathbf{b}'$ given inputs $\{a'_1, a'_2, \dots, a'_n\}$ (we can indicate these with the input vector $\mathbf{a}'$). It turns out that we'll need more than just one input-output pair to train our neural network, so let's define our dataset as the following pair of matrices:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pn} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mp} \end{bmatrix}$$

where $p \geq m, n$. For these matrices, a_{ki} is the value of a_i in the k th training input, and b_{jk} is the value of b_j in the k th training output. We can define each training pair $(\mathbf{a}_k, \mathbf{b}_k)$ as the k th **row** of matrix $\mathbf{A}$ and k th **column** of $\mathbf{B}$.

Assuming that our matrix-vector multiplication from the previous subpart can be written in the form $\mathbf{b} = \mathbf{X}^\top \mathbf{a}$, we want to find the matrix $\mathbf{X}$ that minimizes

$$\arg \min_{\mathbf{X} \in \mathbb{R}^{n \times m}} \|\mathbf{X}^\top \mathbf{A}^\top - \mathbf{B}\|_F^2$$

given our training data $(\mathbf{A}, \mathbf{B})$. The norm $\|\mathbf{M}\|_F$ of a matrix $\mathbf{M}$ is known as the *Frobenius norm* of that matrix. For this problem, it is sufficient to know that $\|\mathbf{M}\|_F^2 = \sum_{i=1}^n \|\mathbf{m}_i\|_2^2$, where $\{\mathbf{m}_1, \mathbf{m}_2, \dots, \mathbf{m}_n\}$ are the rows of $\mathbf{M}$. In other words, the squared Frobenius norm of a matrix is equal to the sum of the squared ℓ_2 norms of its rows.

As it turns out, this is actually the Least Squares problem, but with matrices instead! For this problem, assume that the columns of $\mathbf{A}$ are linearly independent.

- (c) Before we write an expression for the trained value of our entire matrix $\mathbf{X}$, let us consider a simpler version of the problem,

$$\arg \min_{\mathbf{x}_j \in \mathbb{R}^n} \|\mathbf{x}_j^\top \mathbf{A}^\top - \mathbf{b}_j^\top\|^2$$

where $\mathbf{x}_j$ is the j th column of $\mathbf{X}$ and $\mathbf{b}_j^\top$ is the j th row of our output matrix $\mathbf{B}$ (you can verify this by matching up dimensions). Intuitively, this optimization should find the weights $\{x_{1j}, x_{2j}, \dots, x_{mj}\}$ that best map the values of all of the inputs to the output entry b_j of our network. Use Least Squares to find an expression for this optimal vector of weights, $\mathbf{x}_j^*$, in terms of $\mathbf{A}$ and $\mathbf{b}_j$.

Solution: First, because the norm is equivalent between a vector and its transpose, we can rewrite the given optimization problem as

$$\begin{aligned} \arg \min_{\mathbf{x}_j \in \mathbb{R}^n} \|(\mathbf{x}_j^\top \mathbf{A}^\top - \mathbf{b}_j^\top)^\top\|^2 \\ = \arg \min_{\mathbf{x}_j \in \mathbb{R}^n} \|\mathbf{A}\mathbf{x}_j - \mathbf{b}_j\|^2 \end{aligned}$$

By the Least Squares formula, we get that:

$$\mathbf{x}_j^* = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}_j$$

- (d) Now that we have derived a closed-form solution for a particular set of weights $\mathbf{x}_j^*$, derive a closed-form solution for the optimal value of *all* weights, $\mathbf{X}^*$, in terms of $\mathbf{A}$ and $\mathbf{B}$. It may help to consider to property of the Frobenius norm provided above.

Solution: We have derived that for each column of $\mathbf{X}$, the optimal assignment of weights is

$$\mathbf{x}_j^* = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}_j$$

Importantly, these optimal values are independent of the optimal values for the other columns of $\mathbf{X}$. Per the description above regarding the Frobenius norm, we can rewrite our matrix optimization problem as

$$\arg \min_{\mathbf{X} \in \mathbb{R}^{m \times n}} \|\mathbf{X}^\top \mathbf{A}^\top - \mathbf{B}\|_F^2$$

$$= \arg \min_{\mathbf{X} \in \mathbb{R}^{m \times n}} \sum_{j=1}^m \|\mathbf{x}_j^\top \mathbf{A}^\top - \mathbf{b}_j^\top\|_2^2$$

since j th row of $\mathbf{X}^\top \mathbf{A}^\top - \mathbf{B}$ is $\mathbf{x}_j^\top \mathbf{A}^\top - \mathbf{b}_j^\top$. Because the optimal values of each $\mathbf{x}_j^*$ are independent of one another, we can additionally say that the optimal value of this reformulated optimization problem must simply be the $\mathbf{X}$ whose columns are the solutions to the individual Least Squares problem

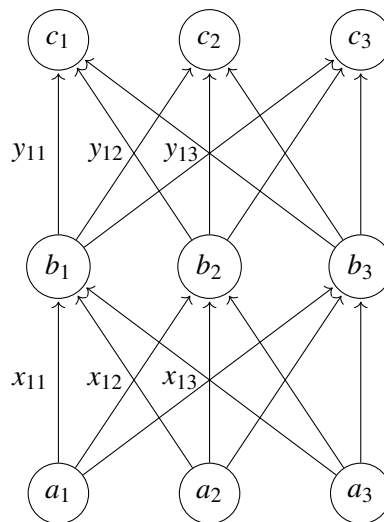
$$\arg \min_{\mathbf{x}_j \in \mathbb{R}^n} \|\mathbf{x}_j^\top \mathbf{A}^\top - \mathbf{b}_j^\top\|^2 \quad \forall j \in \{1, \dots, m\}$$

Similar to part (b), we can observe that the matrix $(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top$ is consistent among the solutions for each column, so we can "stack" the closed-form solutions for each column $\mathbf{x}_j^*$ to get

$$\mathbf{X}^* = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{B}^\top$$

Note that we have $\mathbf{B}^\top$ instead of $\mathbf{B}$ since we are stacking the row vectors $\mathbf{b}_j^\top$ as columns. Using this solution, it turns out that we can directly compute the optimal weights for our neural network by simply computing an expression of our training inputs and outputs!

Finally, consider the *two-layer* linear neural network below.



Observe that it is identical to the single-layer network we have been working with, but now the results $\mathbf{b}$ become the inputs to another single-layer network that outputs $\mathbf{c}$. In this case, the intermediate value $\mathbf{b}$ is known as the *hidden layer*. To train and perform inference using this network, we can simply extend the calculations we performed for the single-layer network.

- (e) Using your result from part (b), express the output of this network, $\mathbf{c}$, as a matrix-vector multiplication. Make sure to define the matrix and vector involved. This is the expression used to perform inference for our two-layer linear network.

Solution: From part (b), we know that the formula for our original single-layer network was

$$\mathbf{b} = \mathbf{X}^\top \mathbf{a}$$

where

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1m} \\ x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

For the second layer of our network, the expression for $\mathbf{c}$ will be the same, but now $\mathbf{b}$ is the input and $\mathbf{c}$ is the output. We can write this as:

$$\mathbf{c} = \mathbf{Y}^\top \mathbf{b}$$

where

$$\mathbf{Y} = \begin{bmatrix} y_{11} & y_{12} & \cdots & y_{1\ell} \\ y_{21} & y_{22} & \cdots & y_{2\ell} \\ \vdots & \vdots & \ddots & \vdots \\ y_{m1} & y_{m2} & \cdots & y_{m\ell} \end{bmatrix}$$

Combining these together, we can say that

$$\mathbf{c} = (\mathbf{Y}^\top \mathbf{X}^\top) \mathbf{a} = (\mathbf{XY})^\top \mathbf{a} = \mathbf{Z}^\top \mathbf{a}$$

where

$$\mathbf{Z} = \mathbf{XY} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1m} \\ x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix} \begin{bmatrix} y_{11} & y_{12} & \cdots & y_{1\ell} \\ y_{21} & y_{22} & \cdots & y_{2\ell} \\ \vdots & \vdots & \ddots & \vdots \\ y_{m1} & y_{m2} & \cdots & y_{m\ell} \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

- (f) Assuming that your equation from the previous subpart was of the form $\mathbf{c} = \mathbf{N}^\top \mathbf{d}$ for some $\mathbf{N}, \mathbf{d}$, write the closed form solution for the optimal matrix $\mathbf{N}^*$ given the input-output data matrices $\mathbf{A}, \mathbf{C}$ akin to the ones described above. What does this imply about the comparison between a single-layer and two-layer linear neural network?

Solution: Pattern-matching to the solution for part (d), we can see that

$$\mathbf{N}^* = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{C}^\top$$

This implies that the trained models for both our single-layer and two-layer linear neural networks are identical!

As it turns out, adding additional layers to purely linear neural networks like the ones explored in this question have no effect on the resulting model. In order to actually benefit from the addition of hidden layers, we must introduce non-linear *activation functions*. These are not in the scope of this class.

To learn more than you ever want to know about neural networks, take CS 189 and CS 182!

2. Eigenvalues and Determinants: Proofs

- (a) Let $\mathbf{A}$ be an invertible matrix. Show that if λ is an eigenvalue of $\mathbf{A}$, then $\frac{1}{\lambda}$ is an eigenvalue of $\mathbf{A}^{-1}$.

Solution: $\mathbf{A}$ is invertible,

$\Rightarrow \mathbf{Ax} = \mathbf{b}$ has a unique solution for all $\mathbf{b}$.

$\Rightarrow \mathbf{Ax} = \mathbf{0}$ has a unique solution.

$\Rightarrow \mathbf{x} = \mathbf{0}$ is the only solution to $\mathbf{Ax} = \mathbf{0}$

$\Rightarrow \mathbf{Ax} = \mathbf{0}$ has no non-zero vectors $\mathbf{x}$ that satisfy it.

Therefore, 0 is not an eigenvalue. Let $\mathbf{v}$ be the eigenvector of $\mathbf{A}$ corresponding to λ .

$$\mathbf{Av} = \lambda \mathbf{v}$$

Since we know that $\mathbf{A}$ is invertible, we can left-multiply both sides by $\mathbf{A}^{-1}$.

$$\mathbf{A}^{-1}\mathbf{Av} = \lambda \mathbf{A}^{-1}\mathbf{v}$$

$$\mathbf{v} = \lambda \mathbf{A}^{-1}\mathbf{v}$$

$$\mathbf{A}^{-1}\mathbf{v} = \frac{1}{\lambda}\mathbf{v}$$

Note that it is important for our eigenvalues to be non-zero, since otherwise $\frac{1}{\lambda} = \frac{1}{0}$ is undefined.

- (b) Given that $\det(\mathbf{A}) = \det(\mathbf{A}^\top)$, show that $\mathbf{A}$ and $\mathbf{A}^\top$ have the same eigenvalues.

Solution: The eigenvalues of $\mathbf{A}$ satisfy $\det(\lambda \mathbf{I} - \mathbf{A}) = 0$. However, because determinants of a matrix and its transpose are equal, then it is the case that

$$\det(\lambda \mathbf{I} - \mathbf{A}) = \det((\lambda \mathbf{I} - \mathbf{A})^\top) = \det(\lambda \mathbf{I}^\top - \mathbf{A}^\top) = \det(\lambda \mathbf{I} - \mathbf{A}^\top).$$

For the choice of λ that makes $\det(\lambda \mathbf{I} - \mathbf{A}) = 0$, we have that $\det(\lambda \mathbf{I} - \mathbf{A}^\top) = 0$ as well. So the eigenvalues of $\mathbf{A}$ are the same as those of $\mathbf{A}^\top$.

- (c) Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$. Assume that $\mathbf{A}$ is invertible, but $\mathbf{B}$ is not invertible. Show that neither $\mathbf{AB}$ nor $\mathbf{BA}$ is invertible.

Solution: There are two ways to show this:

- i. The easiest way to prove this is to use determinants. Since $\mathbf{B}$ is not invertible, $\det(\mathbf{B}) = 0$.

$$\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B}) = \det(\mathbf{A}) \cdot 0 = 0$$

$$\det(\mathbf{BA}) = \det(\mathbf{B})\det(\mathbf{A}) = 0 \cdot \det(\mathbf{A}) = 0$$

Since $\det(\mathbf{AB}) = \det(\mathbf{BA}) = 0$, neither $\mathbf{AB}$ nor $\mathbf{BA}$ is invertible.

- ii. Since $\mathbf{B}$ is not invertible, there must exist a non-zero vector $\mathbf{x}$, such that $\mathbf{Bx} = \mathbf{0}$. We then left-multiply both sides by $\mathbf{A}$.

$$\mathbf{A}(\mathbf{Bx}) = \mathbf{A0} \implies \mathbf{ABx} = \mathbf{0}$$

Since $\mathbf{x}$ is a non-zero vector, $\mathbf{AB}$ has a non-trivial null space, so it is not invertible.

To prove that $\mathbf{BA}$ is not invertible, we know that there exists a vector $\mathbf{y}$, such that $\mathbf{A}^{-1}\mathbf{x} = \mathbf{y} \iff \mathbf{x} = \mathbf{Ay}$ since the inverse $\mathbf{A}^{-1}$ exists. $\mathbf{y}$ is a non-zero vector because $\mathbf{x}$ is a non-zero vector. But for some $\mathbf{x}$ in the null space of $\mathbf{B}$,

$$\mathbf{Bx} = \mathbf{0} \implies \mathbf{B(Ay)} = \mathbf{BAy} = \mathbf{0}$$

Since $\mathbf{y}$ is a non-zero vector, $\mathbf{BA}$ has a non-trivial null space, so it is not invertible as well.

3. Mechanical Diagonalization

All calculations in this problem are intended to be done by hand, but you can use a computer to check your work.

- (a) Diagonalize the matrix $\mathbf{A}$. In other words, compute $\mathbf{V}_\mathbf{A}$, $\mathbf{\Lambda}_\mathbf{A}$, and $\mathbf{V}_\mathbf{A}^{-1}$ such that $\mathbf{A} = \mathbf{V}_\mathbf{A} \mathbf{\Lambda}_\mathbf{A} \mathbf{V}_\mathbf{A}^{-1}$ and the matrix $\mathbf{\Lambda}_\mathbf{A}$ is diagonal with the eigenvalues along the diagonal.

$$\mathbf{A} = \begin{bmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix} \quad (1)$$

Solution: We want to write $\mathbf{A}$ as $\mathbf{A} = \mathbf{V}_\mathbf{A} \mathbf{\Lambda}_\mathbf{A} \mathbf{V}_\mathbf{A}^{-1}$ where $\mathbf{\Lambda}_\mathbf{A}$ is diagonal.

We must first compute the eigenvalues of $\mathbf{A}$. We do so as follows, using the determinant interpretation with the characteristic polynomial:

$$\det(\lambda \mathbf{I} - \mathbf{A}) = 0 \quad (2)$$

$$\det \left(\lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix} \right) = 0 \quad (3)$$

$$\det \left(\begin{bmatrix} \lambda - \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \lambda - \frac{3}{2} \end{bmatrix} \right) = 0 \quad (4)$$

$$\left(\lambda - \frac{3}{2} \right)^2 - \left(-\frac{1}{2} \right)^2 = 0 \quad (5)$$

$$\lambda^2 - 3\lambda + \frac{9}{4} - \frac{1}{4} = 0 \quad (6)$$

$$\lambda^2 - 3\lambda + 2 = 0 \quad (7)$$

$$(\lambda - 1)(\lambda - 2) = 0 \quad (8)$$

$$\lambda = 1, 2 \quad (9)$$

Therefore, $\mathbf{A}$ has eigenvalues $\lambda = 1, 2$.

From here, we can compute the eigenvectors of $\mathbf{A}$. For $\lambda = 1$ we get

$$\text{Null}(\mathbf{I} - \mathbf{A}) = \text{Null} \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix} \right) = \text{Null} \left(\begin{bmatrix} -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} \end{bmatrix} \right) = \text{span} \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) \quad (10)$$

and for $\lambda = 2$ we get

$$\text{Null}(2\mathbf{I} - \mathbf{A}) = \text{Null} \left(\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix} \right) = \text{Null} \left(\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \right) = \text{span} \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) \quad (11)$$

$\mathbf{V}_\mathbf{A}$ is a matrix whose columns are the eigenvectors of $\mathbf{A}$. (Note: the norm of the columns of $\mathbf{V}_\mathbf{A}$ doesn't matter. Why not?) Thus we can write $\mathbf{V}_\mathbf{A}$ and $\mathbf{V}_\mathbf{A}^{-1}$ as

$$\mathbf{V}_\mathbf{A} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \quad \mathbf{V}_\mathbf{A}^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad (12)$$

Thus we can write $\mathbf{A}$ in diagonal form and check it

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix} \quad (13)$$

(b) As with the previous subpart, diagonalize the matrix $\mathbf{B}$.

$$\mathbf{B} = \begin{bmatrix} 1 & 0 \\ -2 & -1 \end{bmatrix} \quad (14)$$

Solution: As with the previous subpart, we must first compute the eigenvalues of $\mathbf{B}$:

$$\det(\lambda \mathbf{I} - \mathbf{B}) = 0 \quad (15)$$

$$\det \left(\lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ -2 & -1 \end{bmatrix} \right) = 0 \quad (16)$$

$$\det \left(\begin{bmatrix} \lambda - 1 & 0 \\ 2 & \lambda + 1 \end{bmatrix} \right) = 0 \quad (17)$$

$$(\lambda - 1)(\lambda + 1) - 0(2) = 0 \quad (18)$$

$$\lambda^2 + \lambda - \lambda - 1 - 0 = 0 \quad (19)$$

$$\lambda^2 - 1 = 0 \quad (20)$$

$$\lambda = 1, -1 \quad (21)$$

Therefore, $\mathbf{B}$ has eigenvalues $\lambda = 1, -1$.

From here, we can compute the eigenvectors.

For $\lambda = 1$,

$$\text{Null}(\mathbf{I} - \mathbf{B}) = \text{Null} \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ -2 & -1 \end{bmatrix} \right) = \text{Null} \left(\begin{bmatrix} 0 & 0 \\ 2 & 2 \end{bmatrix} \right) = \text{span} \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) \quad (22)$$

For $\lambda = -1$,

$$\text{Null}(-\mathbf{I} - \mathbf{B}) = \text{Null} \left(\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ -2 & -1 \end{bmatrix} \right) = \text{Null} \left(\begin{bmatrix} -2 & 0 \\ 2 & 0 \end{bmatrix} \right) = \text{span} \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) \quad (23)$$

Again, $\mathbf{V}_\mathbf{B}$ is a matrix whose columns are the eigenvectors of $\mathbf{A}$. Thus we can write $\mathbf{V}_\mathbf{B}$ and $\mathbf{V}_\mathbf{B}^{-1}$ as

$$\mathbf{V}_\mathbf{B} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \quad \mathbf{V}_\mathbf{B}^{-1} = \frac{1}{1} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \quad (24)$$

Thus we can write $\mathbf{B}$ in diagonal form and check it

$$\mathbf{B} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -2 & -1 \end{bmatrix} \quad (25)$$

(c) Find the matrices $\mathbf{V}_C$ and $\mathbf{\Lambda}_C$ for the diagonalization of the matrix $\mathbf{C} = \mathbf{V}_C \mathbf{\Lambda}_C \mathbf{V}_C^{-1}$ below

$$\mathbf{C} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} & \frac{1}{2} \\ -1 & 1 & 1 \end{bmatrix}$$

given that two of $\mathbf{C}$'s eigenvalues are 2 and 1. Note that you are not required to compute $\mathbf{V}_C^{-1}$.

Solution:

First, we find the missing eigenvalue. One approach is to recognize that the first and third columns of $\mathbf{C}$ are linearly dependent on each other. This means that $\mathbf{C}$ has a nontrivial null space and therefore an eigenvalue of 0. The second approach is to use the property that the trace of a matrix is the sum of the eigenvalues. Recognizing that the trace of $\mathbf{C}$ is 3, the remaining eigenvalue must be 0.

Next, we compute the eigenvectors of $\mathbf{C}$ given the eigenvalues.

For $\lambda = 2$, the eigenvector spans

$$\text{Null} \left(\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} & \frac{1}{2} \\ -1 & 1 & 1 \end{bmatrix} \right) = \text{Null} \left(\begin{bmatrix} \frac{3}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 1 & -1 & 1 \end{bmatrix} \right) = \text{span} \left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right)$$

(If we didn't notice by inspection that the sum of the last two columns is 0, we could have solved it using Gaussian Elimination.)

For $\lambda = 1$, the eigenvector spans

$$\text{Null} \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} & \frac{1}{2} \\ -1 & 1 & 1 \end{bmatrix} \right) = \text{Null} \left(\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ 1 & -1 & 0 \end{bmatrix} \right) = \text{span} \left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right)$$

For $\lambda = 0$, the eigenvector simply spans the nullspace of $\mathbf{C}$.

$$\text{Null} \left(\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} & \frac{1}{2} \\ -1 & 1 & 1 \end{bmatrix} \right) = \text{span} \left(\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right)$$

We now write a matrix $\mathbf{V}$ whose columns are the eigenvectors of $\mathbf{C}$.

$$\mathbf{V}_C = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

As a diagonal matrix with entries equal to the eigenvalues of $\mathbf{C}$, our matrix $\mathbf{\Lambda}$ is

$$\mathbf{\Lambda}_C = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Using these two, we can write $\mathbf{C}$'s diagonal form as

$$\mathbf{C} = \mathbf{V}_C \mathbf{\Lambda}_C \mathbf{V}_C^{-1} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}^{-1}$$

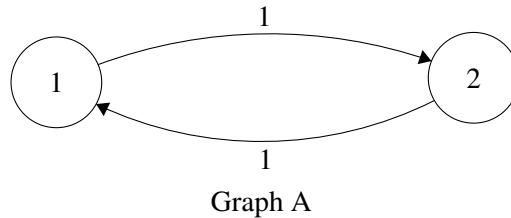
4. PageRank

In this problem, we consider using the adjacency matrices to represent "state transitions" in the setting of web traffic, and how users flow between different websites.

The "connectivity matrix", $\mathbf{C}$, can be constructed using the state transition diagram as follows: entries c_{ji} represent the *proportion* of the people who are at website i that click the link for website j . This is slightly different from the adjacency matrix. Its values, c_{ji} , are the *proportion* of the people who are at website i that click the link for website j . It is associated with a graph that has labels stating the percentage (in our case of people) that is moving from one state to another.

Once computed, an eigenvector with eigenvalue 1 will have values which correspond to the steady-state frequency (i.e. the distribution in the limit as time goes to infinity) for the fraction of people for each webpage. When the elements of this eigenvector are made to **sum to one** (to conserve population), the i^{th} element of the eigenvector will correspond to the fraction of people on the i^{th} website.

- (a) For graph A shown below, what are the steady-state frequencies (i.e. fraction of visitors in steady-state) for the two webpages? Graph A has weights in place to help you construct the connectivity matrix. Remember to ensure that your steady state-frequencies sum to 1 to maintain conservation.



Solution:

To determine the steady-state frequencies for the two pages, we need to find the appropriate eigenvector of the connectivity matrix. In this case, we are trying to determine the proportion of people who would be on a given page at steady state.

The connectivity matrix of graph A:

$$\mathbf{C} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad (26)$$

To determine the eigenvalues of this matrix:

$$\det \left(\begin{bmatrix} \lambda & -1 \\ -1 & \lambda \end{bmatrix} \right) = \lambda^2 - 1 = 0 \quad (27)$$

So we get $\lambda = 1, -1$.

The steady state vector is the eigenvector that corresponds to $\lambda = 1$. To find the eigenvector,

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \quad (28)$$

The sum of the values of the vector should equal 1 since the number of people is conserved, so our conditions are:

$$\begin{aligned} v_1 + v_2 &= 1 \\ v_1 &= v_2 \end{aligned}$$

The steady-state frequency eigenvector is $\begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$ and each webpage has a steady-state frequency of 0.5.

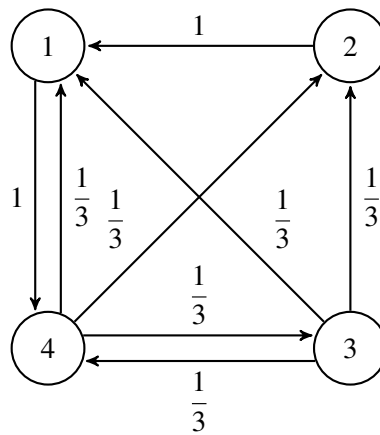
- (b) For graph B, what are the steady-state frequencies for the webpages? You may use IPython and the Numpy command `numpy.linalg.eig` for this. Graph B is shown below, with weights in place to help you construct the connectivity matrix.

Hint: `numpy.linalg.eig` returns eigenvectors and eigenvalues. The eigenvectors are arranged in a matrix in column-major order. In other words, given eigenvectors

$$\mathbf{v}_1 = \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} \text{ and } \mathbf{v}_2 = \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix}$$

NumPy will return:

$$\begin{bmatrix} v_{11} & v_{21} \\ v_{12} & v_{22} \end{bmatrix} \quad (29)$$



Graph B

Solution:

To determine the steady-state frequencies, we need to create the connectivity matrix $\mathbf{C}$ first.

$$\mathbf{C} = \begin{bmatrix} 0 & 1 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{3} \\ 1 & 0 & \frac{1}{3} & 0 \end{bmatrix}$$

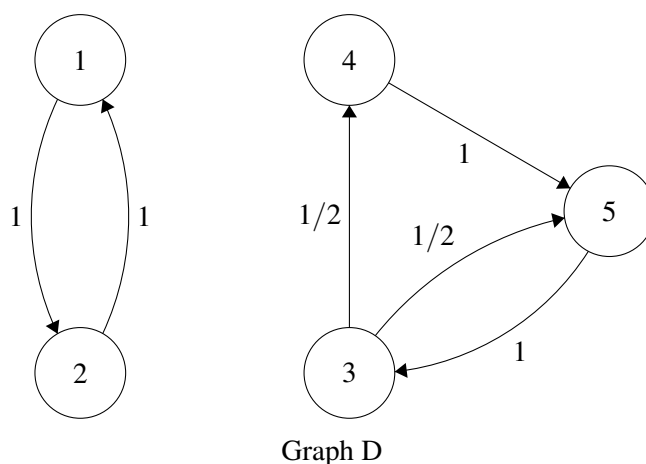
One possible eigenvector associated with eigenvalue 1 is $[-0.61 \quad -0.31 \quad -0.23 \quad -0.69]^\top$ (found using IPython). Scaling by

$$\frac{1}{(-0.61 + (-0.31) + (-0.23) + (-0.69))}$$

so the elements sum to 1, we get $\left[\frac{1}{3} \quad \frac{1}{6} \quad \frac{1}{8} \quad \frac{3}{8}\right]^\top$

These are the steady-state frequencies for the pages.

- (c) Graph D with weights in place is shown below. Find the eigenspace that corresponds to the steady-state for graph D. How many independent systems (disjoint sets of webpages) are there in graph D versus in graph B? What is the dimension of the eigenspace corresponding to the steady-state for graph D? You may use IPython to compute the eigenvalues and eigenvectors again.

**Solution:**

The connectivity matrix for graph D is

$$\mathbf{C} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 1 & 0 \end{bmatrix}$$

Using IPython, we find that the eigenspace associated with $\lambda = 1$ is spanned by the vectors

$$\begin{bmatrix} 0 \\ 0 \\ 0.4 \\ 0.2 \\ 0.4 \end{bmatrix}, \begin{bmatrix} 0.5 \\ 0.5 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

While any linear combination of these vectors is an eigenvector, these two particular vectors have a nice interpretation.

The first eigenvector describes the steady-state frequencies for the last three webpages, and the second vector describes the steady-state frequencies for the first two webpages. This makes sense since there are essentially “two internets”, or two disjoint sets of webpages. Surfers cannot transition between the two, so you cannot assign steady-state frequencies to webpage 1 and webpage 2 relative to the rest. This is why the eigenspace corresponding to the steady-state has dimension 2.

Assuming that each set of steady-state frequencies needs to add to 1, the first assigns steady-state frequencies of 0.4, 0.2, 0.4 to webpage 3, webpage 4, and webpage 5, respectively. The second assigns steady-state frequencies of 0.5 to both webpage 1 and webpage 2.

5. Short Proofs

Let's define $\mathbf{S} = \mathbf{A}^\top \mathbf{A}$, where $\mathbf{A} \in \mathbb{R}^{m \times n}$ is an arbitrary real $m \times n$ matrix. $\mathbf{V} = [\mathbf{v}_1 \cdots \mathbf{v}_n]$ is the matrix of normalized eigenvectors of $\mathbf{S}$ with eigenvalues $\lambda_1, \dots, \lambda_n$.

- (a) Let $\mathbf{v}_1, \dots, \mathbf{v}_r$ be those normalized eigenvectors that correspond to non-zero eigenvalues (i.e. $\|\mathbf{v}_i\| = 1$ and $\lambda_i \neq 0$ for $i = 1, \dots, r$). Show that $\|\mathbf{A}\mathbf{v}_i\|^2 = \lambda_i$.

Solution: We know that $\mathbf{v}_i$ is an eigenvector of $\mathbf{S}$, so we can start with:

$$\begin{aligned}\mathbf{S}\mathbf{v}_i &= \lambda_i \mathbf{v}_i \\ \mathbf{v}_i^\top \mathbf{S}\mathbf{v}_i &= \lambda_i \mathbf{v}_i^\top \mathbf{v}_i. \\ \mathbf{v}_i^\top \mathbf{S}\mathbf{v}_i &= \lambda_i \|\mathbf{v}_i\|^2.\end{aligned}$$

On the second step, we multiplied both sides with $\mathbf{v}_i^\top$. Since $\|\mathbf{v}_i\|^2 = 1$ the right hand side will simplify to λ_i . By substituting $\mathbf{S} = \mathbf{A}^\top \mathbf{A}$:

$$\begin{aligned}\mathbf{v}_i^\top \mathbf{A}^\top \mathbf{A}\mathbf{v}_i &= \lambda_i \\ (\mathbf{A}\mathbf{v}_i)^\top \mathbf{A}\mathbf{v}_i &= \lambda_i \\ \|\mathbf{A}\mathbf{v}_i\|^2 &= \lambda_i.\end{aligned}$$

- (b) Following the assumptions in part (a), show that $\mathbf{A}\mathbf{v}_i$ is orthogonal to $\mathbf{A}\mathbf{v}_j$.

Hint: It may help to recall the Spectral Theorem.

Solution: To show that two vectors are orthogonal we have to take the inner product and show that it is zero:

$$(\mathbf{A}\mathbf{v}_i)^\top \mathbf{A}\mathbf{v}_j = \mathbf{v}_i^\top \mathbf{A}^\top \mathbf{A}\mathbf{v}_j = \mathbf{v}_i^\top \mathbf{S}\mathbf{v}_j.$$

We know that $\mathbf{S}\mathbf{v}_j = \lambda_j \mathbf{v}_j$. Therefore,

$$\mathbf{v}_i^\top \mathbf{S}\mathbf{v}_j = \lambda_j \underbrace{\mathbf{v}_i^\top \mathbf{v}_j}_{=0(\text{since } \mathbf{v}_i \perp \mathbf{v}_j)} = 0.$$

- (c) Show that if $\mathbf{V} \in \mathbb{R}^{n \times n}$ is an orthonormal square matrix, then $\|\mathbf{x}\|^2 = \|\mathbf{V}\mathbf{x}\|^2$ for all $\mathbf{x} \in \mathbb{R}^n$.

Hint: Write the norm as an inner product instead of trying to do this elementwise.

Solution:

$$\|\mathbf{V}\mathbf{x}\|^2 = \langle \mathbf{V}\mathbf{x}, \mathbf{V}\mathbf{x} \rangle = \mathbf{x}^\top \mathbf{V}^\top \mathbf{V}\mathbf{x} = \mathbf{x}^\top \mathbf{I}_{n \times n} \mathbf{x} = \|\mathbf{x}\|^2.$$

6. Homework Process and Study Group

Whom did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

Solution:

I first worked by myself for 2 hours, but got stuck on problem 5. Then I met with my study group.

XYZ played the role of facilitator ... etc. We were still stuck on problem 5 so we went to office hours to talk about the problem.

Then I went to homework party for a few hours, where I finished the homework.